

MODULE 1

INTRODUCTION TO SIEMENS PLC

PLC Siemens TIA Portal Course

Module 1: Introduction to Siemens PLC

Author: AUTOMATION AND ROBOTICS ENGINEER

Level: Basic

Software: Siemens TIA Portal

Language: Ladder Diagram (LAD)

1. WHAT IS A PLC?

A PLC (Programmable Logic Controller) is an industrial electronic device designed to automate machines and processes.

EXAMPLE OF A SIEMENS S7-1200 PLC.



PLCs replace old electrical relay systems, allowing control of motors, sensors, valves, alarms, and industrial systems through programming.

PLCs are used in:

- Industrial plants
- Production lines
- Transport systems
- Automatic machinery
- Food processing
- Pumping systems
- Building automation

MAIN FUNCTIONS OF A PLC

- Read input signals
- Process control logic
- Activate outputs
- Monitor processes
- Control timing and sequences
- Communicate with other devices

MAIN PARTS OF A PLC

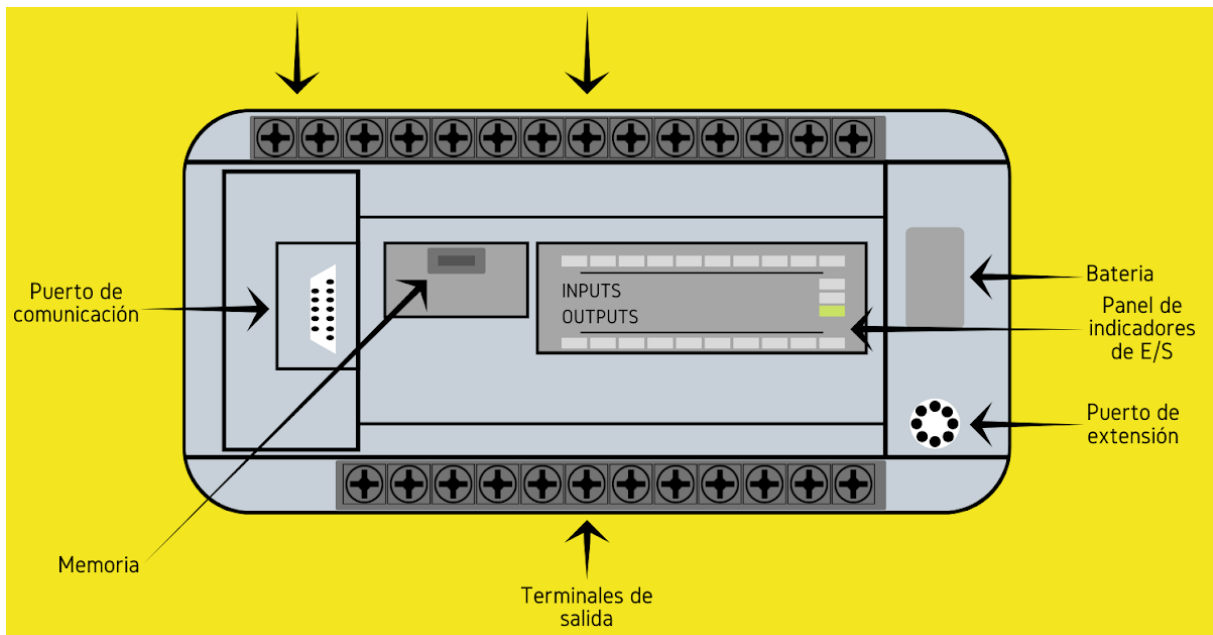


Diagram labels (original graphic): Puerto de comunicación = Communication port · Memoria = Memory · Bateria = Battery · Panel de indicadores de E/S = I/O indicator panel · Puerto de extensión = Extension port · Terminales de salida = Output terminals.

2. SIEMENS TIA PORTAL



TIA Portal (Totally Integrated Automation Portal) is Siemens' industrial programming software.

With TIA Portal it is possible to:

- Program Siemens PLCs
- Design HMIs
- Configure industrial networks
- Diagnose faults
- Simulate industrial processes

3. INPUTS AND OUTPUTS

Inputs and outputs are fundamental in any PLC system.

DIGITAL INPUTS

Inputs receive signals from external devices.

Examples:

- Push buttons
- Inductive sensors
- Photoelectric sensors
- Limit switches
- Emergency stop buttons

ADDRESS EXAMPLE

Address	Description
I0.0	START button
I0.1	STOP button
I0.2	Sensor
I0.3	Emergency stop

DIGITAL OUTPUTS

Outputs activate industrial devices.

Examples:

- Motors
- Pilot lights
- Contactors
- Solenoid valves
- Alarms

ADDRESS EXAMPLE

Address	Description
Q0.0	Motor
Q0.1	Green light
Q0.2	Red light
Q0.3	Fan

4. NO AND NC CONTACTS

Ladder programming uses contacts to represent logic conditions.

NO CONTACT (NORMALLY OPEN)

The NO contact is activated when the signal is true.

Representation:

-- | | --

Example:

If the START button is pressed, the contact allows logic power to flow.

NC CONTACT (NORMALLY CLOSED)

The NC contact is active while the signal is false.

Representation:

-- | / | --

Example:

A STOP button normally uses an NC contact to break the circuit when pressed.

5. LADDER COILS

Coils represent outputs or internal memories.

Representation:

-- () --

Example:

I0.0 ---- (Q0.0)

If I0.0 is true, then Q0.0 is activated.

6. LADDER MINI EXERCISE

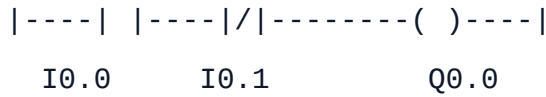
OBJECTIVE

Turn on a motor using a START button and stop it with a STOP button.

ADDRESSES

Address	Function
I0.0	START
I0.1	STOP
Q0.0	MOTOR

LADDER LOGIC



OPERATION

- START activates the motor
- STOP deactivates the motor
- The motor runs as long as there is no stop condition

7. PLC SCAN CYCLE

The PLC works in extremely fast cycles called SCAN.

The cycle is divided into:

- Reading inputs
- Executing the program
- Updating outputs
- Continuous repetition

This process occurs in milliseconds.

8. ADVANTAGES OF PLCs

- High reliability
- Easy maintenance
- Flexible programming
- Less wiring
- Fast diagnostics

- Integration with industrial networks
- Advanced automation

9. FINAL SUMMARY

In this module we learned:

- What a PLC is
- How a Siemens PLC works
- What inputs and outputs are
- Basic TIA Portal concepts
- NO and NC contacts
- Ladder coils
- Basic automation logic

This knowledge is the fundamental basis for beginning to program real industrial systems using Siemens TIA Portal.

END OF MODULE 1

AUTOMATION STUDIO X

PLC Siemens TIA Portal Training System